Spin Ice and Kagome Magnets on Every Stacking

Loops, kagome planes and the Hexagonality Law

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The ideas and names are DICTO's. Computation and proof with Claude (Anthropic).

Version 2 states the Loop Equality as a theorem, proved in #102 (DICTO DOI 10.5281/zenodo.24037664), adds that by its Corollary 2 the pyrochlore network's loops do not depend on the stacking either, and credits the closed-walk equality to Nagle (1966) and Thorpe (1972).

Abstract. Two families of magnets are built from corner-sharing triangles and tetrahedra: spin ice, whose magnetic atoms sit on the pyrochlore network of DICTO Janus c/h Pyrochlore (#60), and the kagome antiferromagnets Mn_3X, whose manganese atoms form kagome nets. Both are close-packed layers, so both can stack cubic, hexagonal or any mixture. We count exactly what changes with the share h of hexagonal layers. On the pyrochlore network of any stacking, three quarters of the sites lie in kagome layers and one quarter in triangular layers; per tetrahedron the boat-shaped six-loops number 3h/2 and the chair-shaped $2 - 3h/2$. Yet the closed loops of the network do not see the stacking at all: on five stackings the numbers of closed non-backtracking loops of every length from 6 to 34 bonds are identical, while the shells of neighbours differ by up to 5%. We call this the Loop Equality; computed here through length 34, it is proved for every length, at every site and on every stacking in #102 (DICTO DOI 10.5281/zenodo.24037664). It gives a counting reason why the frozen-in entropy of spin ice and of water ice is the same for cubic and hexagonal stacking, as recent tensor-network work finds, and predicts the same value for every stacking, the disordered ice of clouds included. In the kagome magnets each manganese atom lies on exactly $1 + 3(1 - h)$ kagome planes: four in cubic Mn_3Ir, kagome in all directions, one in hexagonal Mn_3Sn, kagome in layers only.

1. Triangles, tetrahedra and stacking

In spin ice ($Dy_2Ti_2O_7$, $Ho_2Ti_2O_7$) the magnetic atoms sit at the corners of tetrahedra that touch only at corners: the pyrochlore network. Each tetrahedron must have two spins pointing in and two out, the same rule as the hydrogen atoms of water ice, and like ice the material keeps a frozen-in entropy at the lowest temperatures (Bramwell and Gingras); its defects behave as magnetic monopoles (Castelnovo, Moessner and Sondhi). In a field along the stacking axis the spins of the triangular layers are held and the kagome layers carry the disorder (Isakov et al.); ice on a single kagome layer has its own exact entropy (Lin and Tang). The pyrochlore network is the network of bond midpoints of a diamond-like crystal (#60), so it follows the stacking of the diamond layers: cubic, hexagonal or any mixture, as in #73. In the kagome antiferromagnets Mn_3X the manganese atoms make kagome nets of corner-sharing triangles inside close-packed layers, the same AX_3 layers as the hexagonal perovskites of #72; Mn_3Ir and Mn_3Pt stack them cubic, Mn_3Sn and Mn_3Ge hexagonal. Their

spins on the kagome triangles give a large anomalous Hall effect without net magnetisation, found at room temperature in Mn_3Sn (Nakatsuji, Kiyohara and Higo; reviewed by Šmejkal et al.). DICTO chose these two families as the last materials study.

2. The pyrochlore network on any stacking

> **Theorem 1.** On the pyrochlore network of any close-packed stacking with hexagonality h: (i) there are two sites per atom of the parent network, one quarter in triangular layers and three quarters in kagome layers; (ii) per tetrahedron, h/2 of the corner-sharing pairs are eclipsed (mirror images across the shared corner) and the rest staggered (inversion images).

Proof. Each site is the midpoint of one bond; there are two bonds per atom. The bond along the stacking axis, one per two atoms, has its midpoint in a triangular layer; the three slanted bonds have theirs in kagome layers. Two tetrahedra sharing a corner are the tetrahedra of the two atoms of one bond; they are eclipsed exactly when the bond is eclipsed, which by #73 happens for h/2 bonds per atom. ∎

The loops of the network change shape with the stacking. Counted from explicit coordinates on 3C, 2H, 4H, 6H and 9R, each tetrahedron has 3h/2 boat-shaped six-loops and 2 − 3h/2 chair-shaped ones, two in all, and three eight-loops, the same in every stacking.

3. The Loop Equality

A closed non-backtracking loop is a walk along the network that returns to its start and never turns straight back. These loops carry the corrections to Pauling's estimate of the ice-rule entropy. We counted them on 3C, 2H, 4H, 6H and 9R, from one column of sites in a cell 36 by 36 sites wide and at least 36 layers high, so that no loop up to length 34 can wrap round the cell, with every bond checked against its length.

> **Loop Equality (computed here; a theorem since #102).** On every close-packed stacking, the number of closed non-backtracking loops of each length per site is the same. Computed for all even lengths from 6 to 34 on the five stackings: 24, 48, 480, 3432, 25704, 194256, 1509072, 11820000, …, 27416593190640, identical to the last digit.

The networks themselves differ. The numbers of atoms 3, 4, 5 and 6 bonds away are 24, 42, 64, 92 for the cubic stacking and 25, 44, 67, 96 for the hexagonal, the known values for diamond and lonsdaleite, with 4H in between (Figure 1a). The shells see the stacking from three bonds on; the loops do not see it up to 34.

Entropy. Pauling's estimate gives the same residual entropy, (1/2) ln(3/2) per spin in spin ice, for every stacking. Onsager proved that the hexagonal value is at least the cubic one; Nagle's series found them equal within its precision, and Xu, Lin and Zhang's tensor-network computation finds w = 1.5074584 for both, as numerical evidence that they are equal. The Loop Equality gives a counting reason: the loops that correct Pauling's estimate are the same on both, through loops of 34 bonds. It also extends the statement from the two pure stackings to all of them: to the extent that the entropy is set by these loops, spin ice and water ice have the same frozen-in entropy in every stacking, stacking-disordered ice included. In the language of the Hexagonality Law, this is the law with zero slope.

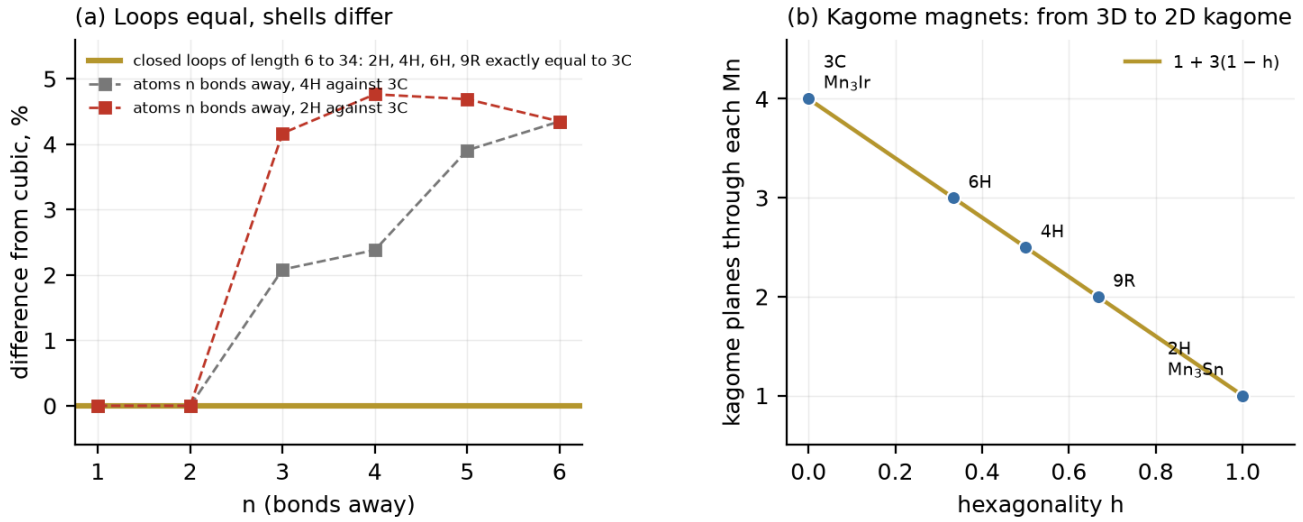


Figure 1. (a) The closed non-backtracking loops of lengths 6 to 34 are exactly equal on 3C, 2H, 4H, 6H and 9R (gold line at zero difference), while the numbers of atoms n bonds away differ by up to 5% from the cubic values for 4H and 2H. (b) Kagome planes through each manganese atom of the A_3B magnets against the hexagonality: $1 + 3(1 - h)$, from four in cubic Mn_3Ir to one in hexagonal Mn_3Sn, exact on all five stackings.

4. Kagome magnets

In Mn_3X each close-packed layer holds manganese on a kagome net and X on the remaining quarter of the sites, and the layers stack so that no two X atoms touch. We counted from explicit coordinates on 3C, 2H, 4H, 6H and 9R. Many things do not change: each manganese has 8 manganese and 4 X neighbours, 2 triangles of manganese in its layer and 6 tilted ones, and one third of an octahedral hole surrounded by manganese only. What changes is how the tilted triangles join.

> **Theorem 2.** In an A_3B close-packed stacking with hexagonality h, each A atom lies on $1 + 3(1 - h)$ kagome planes: always its own layer, and three tilted planes if its layer is cubic, none if it is hexagonal.

Proof. A kagome plane through the atom needs two triangles of the plane meeting at the atom, one on each side. For a tilted plane, one triangle reaches into the layer below and the other into the layer above, and the two lie in one plane exactly when the atoms above are the images of the atoms below under inversion through the atom. That is so when the layers below and above sit on different positions, a cubic layer, for each of the three tilted directions; when they sit on the same position, a hexagonal layer, the images are mirror images and no tilted pair is coplanar. Averaging over the layers gives $1 + 3(1 - h)$. ∎ The count was checked on all five stackings ($1 + 3(1 - h) = 4, 3, 2.5, 2, 1$ for 3C, 6H, 4H, 9R, 2H).

So the law counts the dimension of the kagome network. In cubic Mn_3Ir every manganese lies on four kagome planes, one in each of the four (111) directions, as Chen, Niu and MacDonald describe; in hexagonal Mn_3Sn on one, the basal plane. The intermediate stackings, such as the 4H ($D0_{24}$) type, would have kagome planes in between, 2.5 per atom on average: half the atoms with four, half with one. The Swap Rule of #61 and #64 applies to these layers unchanged, since they are the close-packed AX_3 layers treated there and in #72; we do not test it again.

5. Where the results stop

(1) The Loop Equality was computed here through length 34 on five stackings; #102 proves it for all lengths and all stackings, periodic or not. The same equality for closed walks, which may turn straight back, was known before: Nagle (1966) for ice, and Thorpe (1972, J. Math. Phys. 13, 294–299) for all periodic polytypes. (2) The step from equal loops to equal entropy is not exact: the corrections to Pauling's estimate come from clusters of loops, not only from loop counts; the agreement with the tensor-network values supports it but does not prove it. (3) Theorem 2 counts geometry only; which kagome planes carry the spin order and the Hall effect depends on the magnetic structure, and for Mn_3Sn the basal-plane Hall response is forbidden by symmetry (Liu et al.).

6. Conclusion

The Hexagonality Law reaches magnets in two ways. The kagome planes through each manganese atom of the Mn_3X magnets fall on its straight line, $1 + 3(1 - h)$, from three-dimensional kagome in Mn_3Ir to two-dimensional in Mn_3Sn. The loops of the tetrahedral network do not depend on the stacking at all (through length 34 here, for every length by #102), nor, by #102's Corollary 2, do those of the pyrochlore network, so the frozen-in entropy of spin ice and ice is the same for every stacking: the law with zero slope. This study closes the line of work on materials.

7. Prior art and credit

The ice rule and its entropy estimate are Pauling's; the inequality between the hexagonal and cubic entropies Onsager's; the series Nagle's; the numerical values Herrero and Ramírez's and Xu, Lin and Zhang's, who give evidence that the two are equal; the kagome description of Mn_3Ir is Chen, Niu and MacDonald's. The pyrochlore network as bond midpoints of a stacked diamond network is DICTO Janus c/h Pyrochlore (#60). Stated here: the counts of Theorems 1 and 2 with proofs, the loop shapes, and the Loop Equality as computed. We found no earlier statement that diamond-like networks of different stacking have equal counts of non-backtracking loops; for closed walks the equality is Nagle's (1966) and Thorpe's (1972), as #102 records. DICTO chose the two families and set the end of the materials line; the computations are joint work.

> **Sources checked** (28 October 2026). Read: X.-Z. Xu, T.-Y. Lin and G.-M. Zhang, Equivalence of residual entropy of hexagonal and cubic ices from tensor network methods, arXiv 2511.22477 v3 (pp. 1, 7–8). Search summaries, not read in full: C. P. Herrero and R. Ramírez, Configurational entropy of hydrogen-disordered ice polymorphs, arXiv 1406.5929; arXiv 2404.13897 (transfer matrices, Onsager's inequality as reported in Nagle's thesis); arXiv 2009.11591; H. Chen, Q. Niu and A. H. MacDonald, Anomalous Hall effect arising from noncollinear antiferromagnetism, arXiv 1309.4041; J. Kübler and C. Felser, arXiv 1410.5985 (Mn_3Sn, Mn_3Ge); Y. Liu et al., Giant full-space anomalous Hall effect induced by non-coplanar spin state in Mn-rich Mn_3Sn, arXiv 2603.13718 (abstract and reference list read). Cited as given in the reference list of Xu et al., not read: L. Pauling, J. Am. Chem. Soc. 57, 2680 (1935); J. F. Nagle, J. Math. Phys. 7, 1484 (1966); E. H. Lieb, Phys. Rev. Lett. 18, 692 (1967); K. Y. Lin and D. L. Tang, J. Phys. A 9, 1101 (1976); S. V. Isakov, K. S. Raman, R. Moessner and S. L. Sondhi, Phys. Rev. B 70, 104418 (2004); C. P. Herrero and R. Ramírez, Chem. Phys. Lett. 568–569, 70 (2013). Cited as given in the reference list of Liu et al., not read: S. Nakatsuji, N. Kiyohara and T. Higo, Large anomalous Hall effect in a non-collinear antiferromagnet at room temperature, Nature 527, 212 (2015); L. Šmejkal, A. H. MacDonald, J. Sinova, S. Nakatsuji and T. Jungwirth, Anomalous Hall antiferromagnets, Nat. Rev. Mater. 7, 482 (2022). Records checked on arXiv (title, authors, DOI), not read: S. T. Bramwell and M. J. P. Gingras, Spin ice state in

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Data and code. Study 2026-10-28-pyrochlore-kagome (pyro_counts.py, loops2.py, coordseq.py, a3b_counts.py, fig74.py, with outputs; loops.py kept as the superseded small-cell run), attached to this record. References: DICTO, The Rule of Halves (#57), DICTO DOI 10.5281/zenodo.23647898; DICTO Janus c/h Pyrochlore (#60), 10.5281/zenodo.23678750; The Swap Rule (#61), 10.5281/zenodo.23682596; The Swap Rule in Crystals (#64), 10.5281/zenodo.23709753; The Focus (#66), 10.5281/zenodo.23717687; The Hexagonality Law in Real Materials IV (#72), 10.5281/zenodo.23728287; V (#73), 10.5281/zenodo.23728921.